

Networks and Systems Management



Routing
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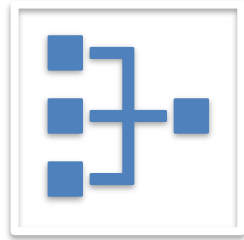
Introduction to Routing



What is Routing?

Process of forwarding data packets from source to destination

Uses routing tables to determine the best path



Types of Routing:

Static Routing (Manually configured paths)

Dynamic Routing (Automated path selection using protocols)



Routing Basics



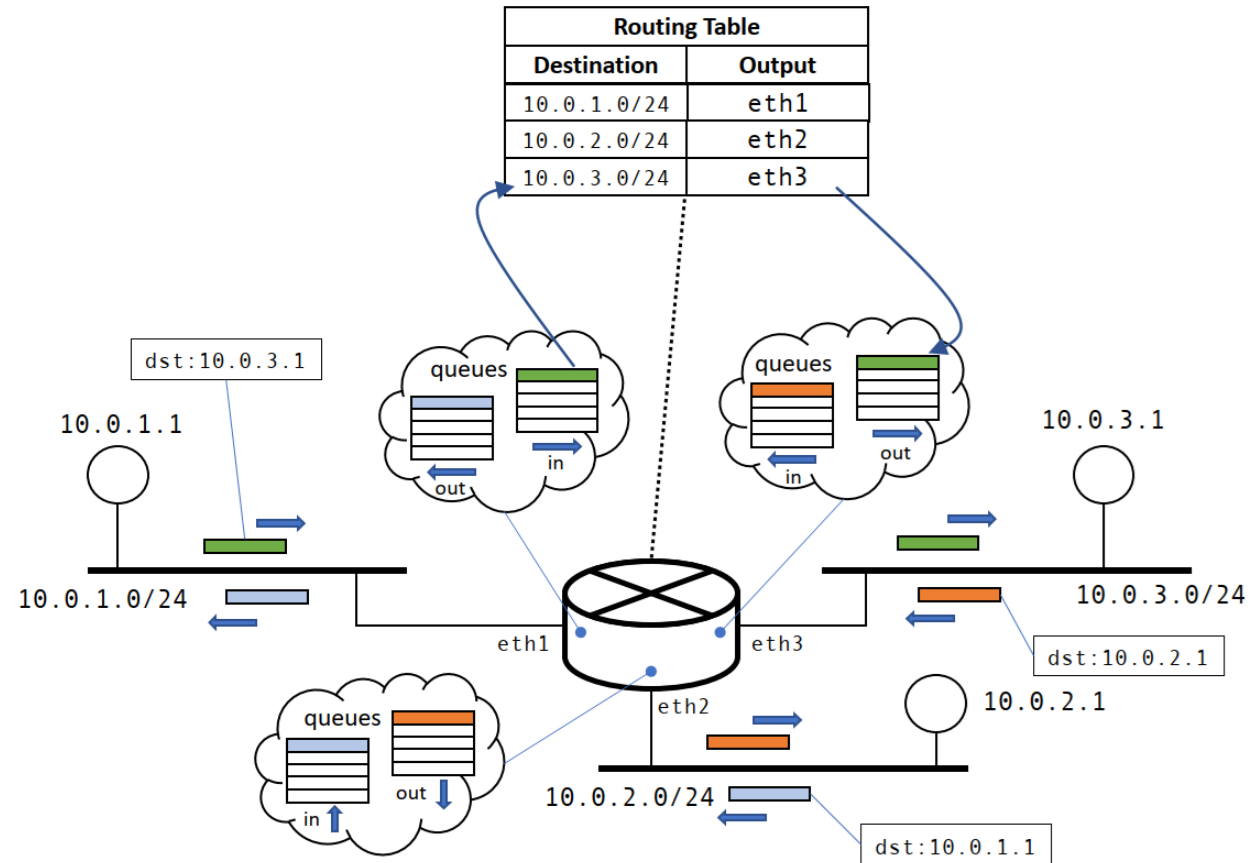
Forwarding
between Interfaces



Input/output
queues

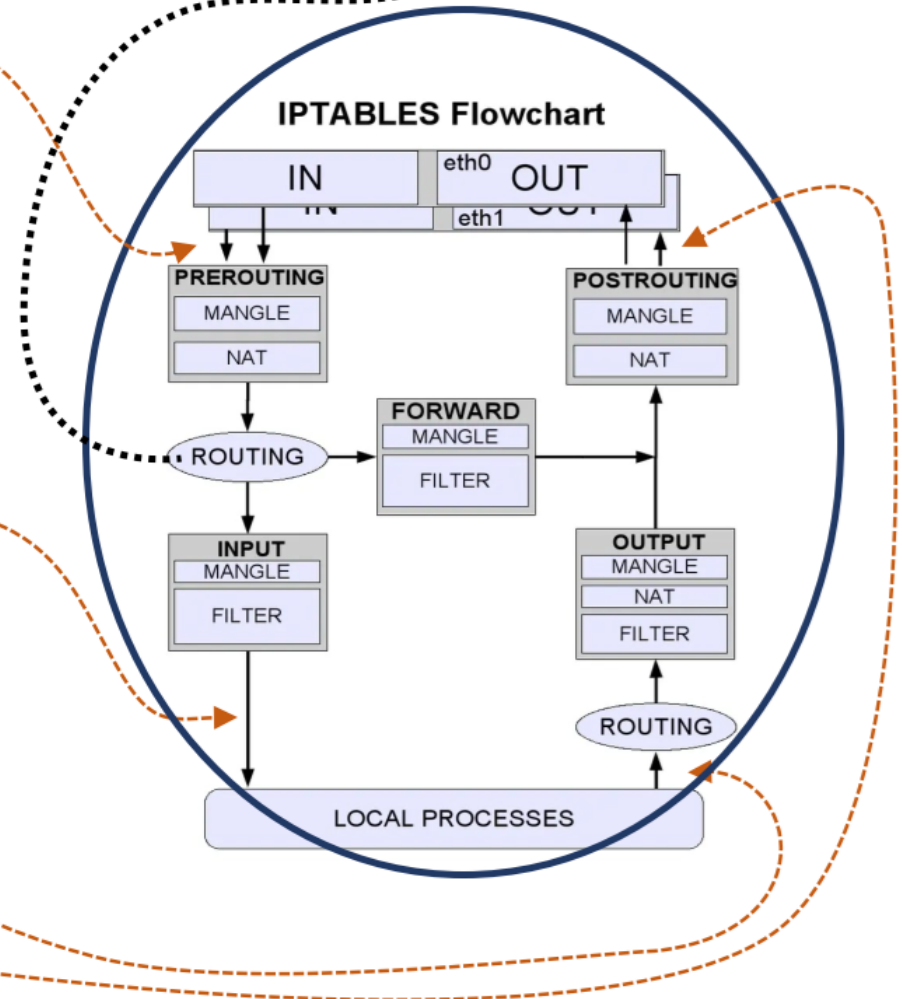
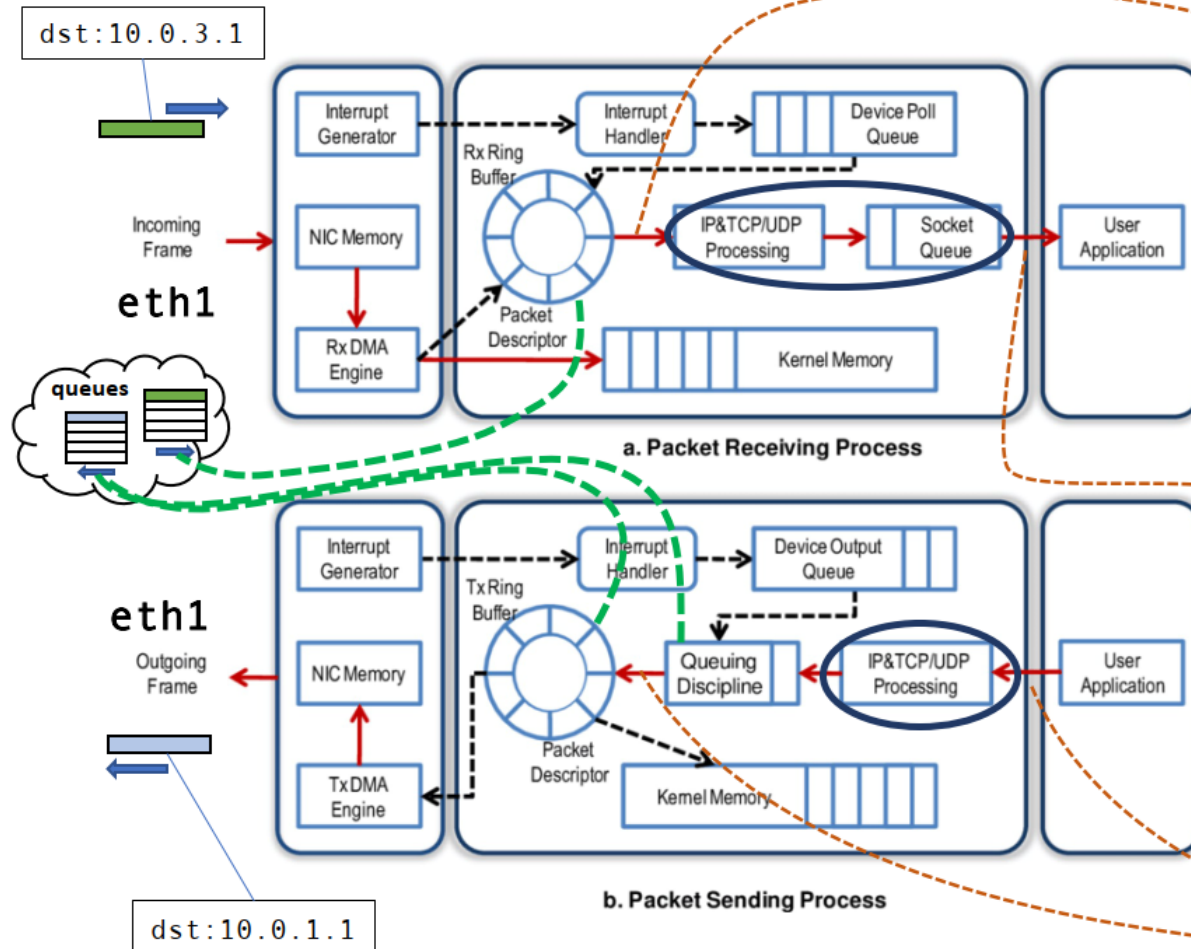


Routing table

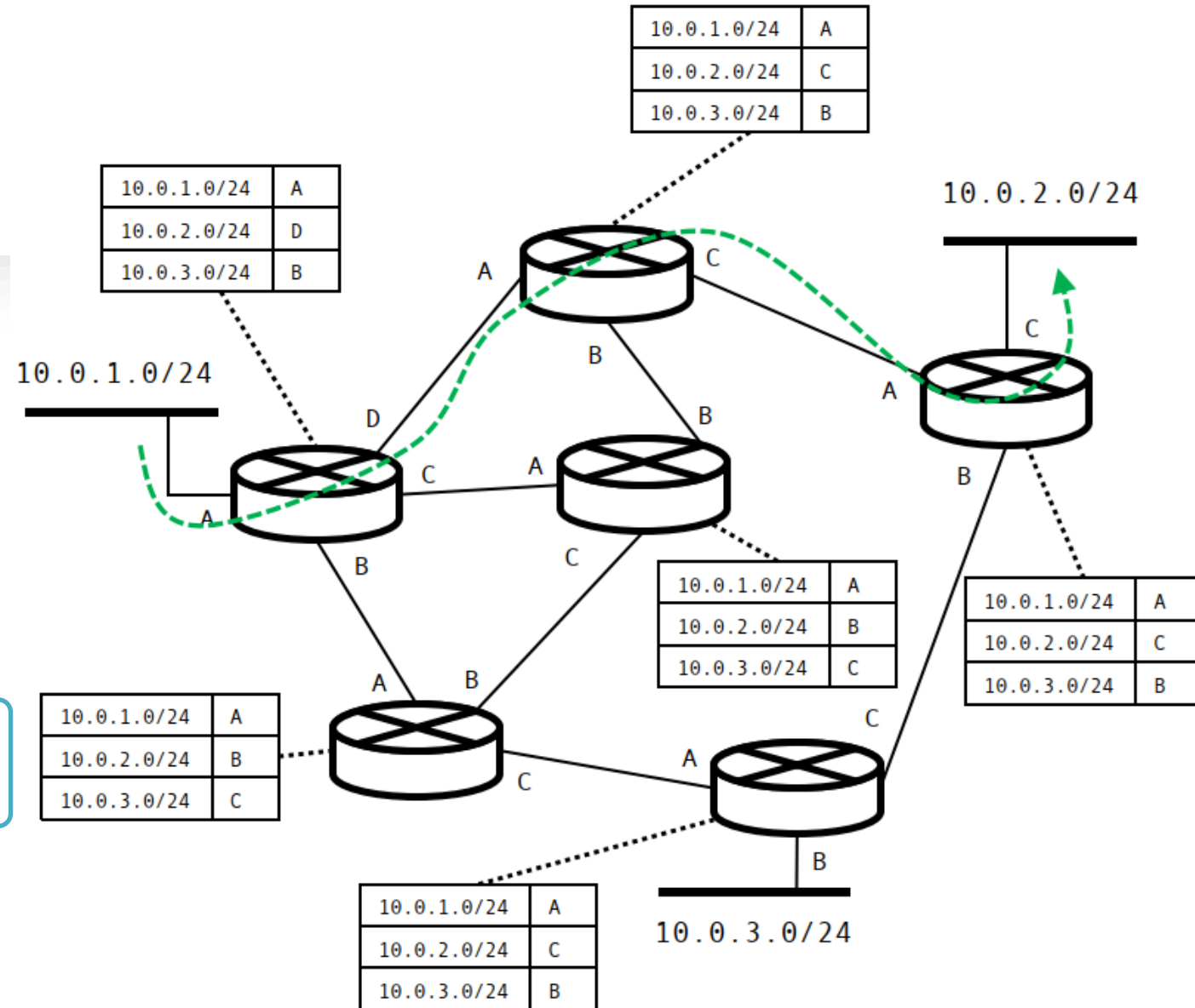
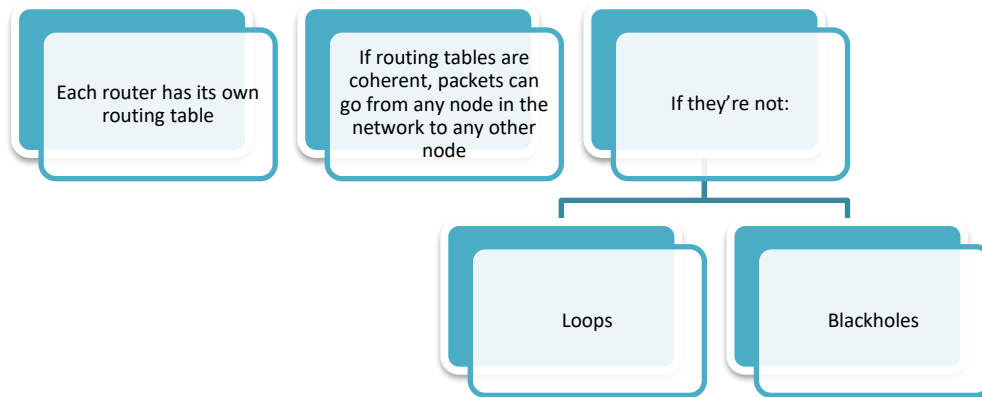


Inside the router

Routing Table	
Destination	Output
10.0.1.0/24	eth1
10.0.2.0/24	eth2
10.0.3.0/24	eth3



Routing path



Routing topology

Each router can sense if their links are up or down

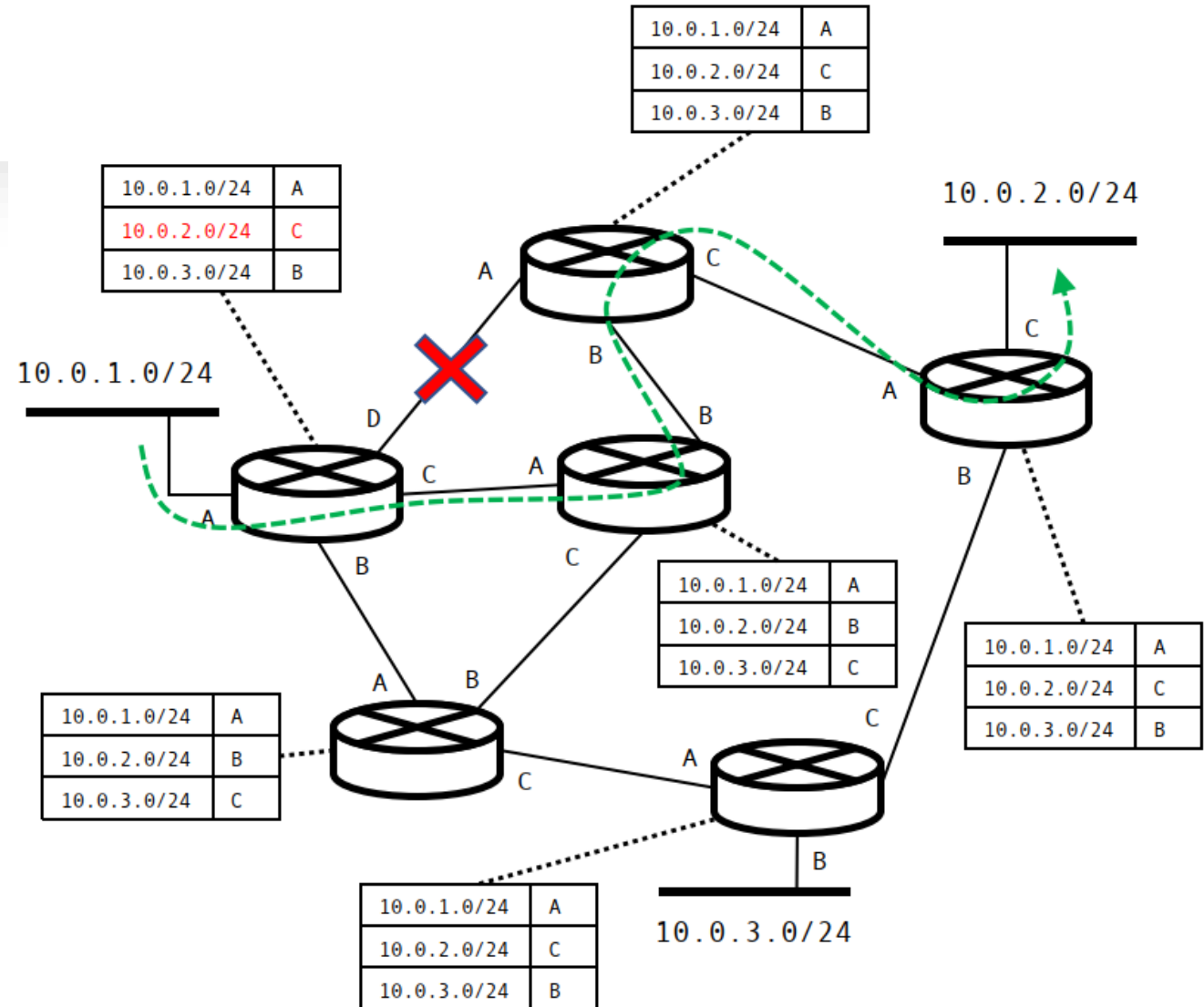
- Or if new networks are added

Automatically update routing tables

- New path bypasses down link

Based on some objective to optimize:

- Smallest number of hops
- Smallest delay
- Largest throughput
- Balance load on links
- Administrative distance (...)



Centralized routing

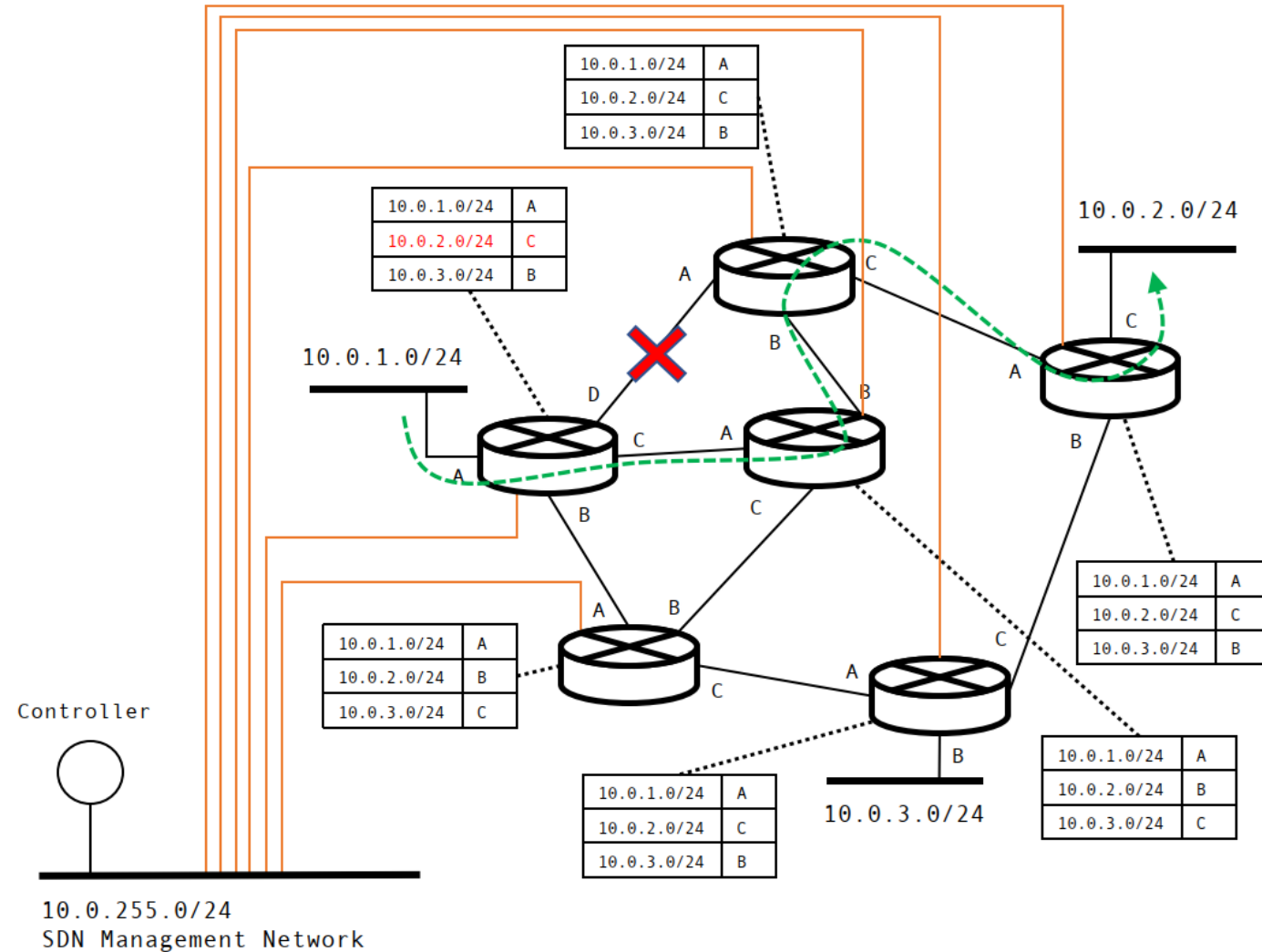
Nodes report link status to central location

Central location runs optimization algorithm

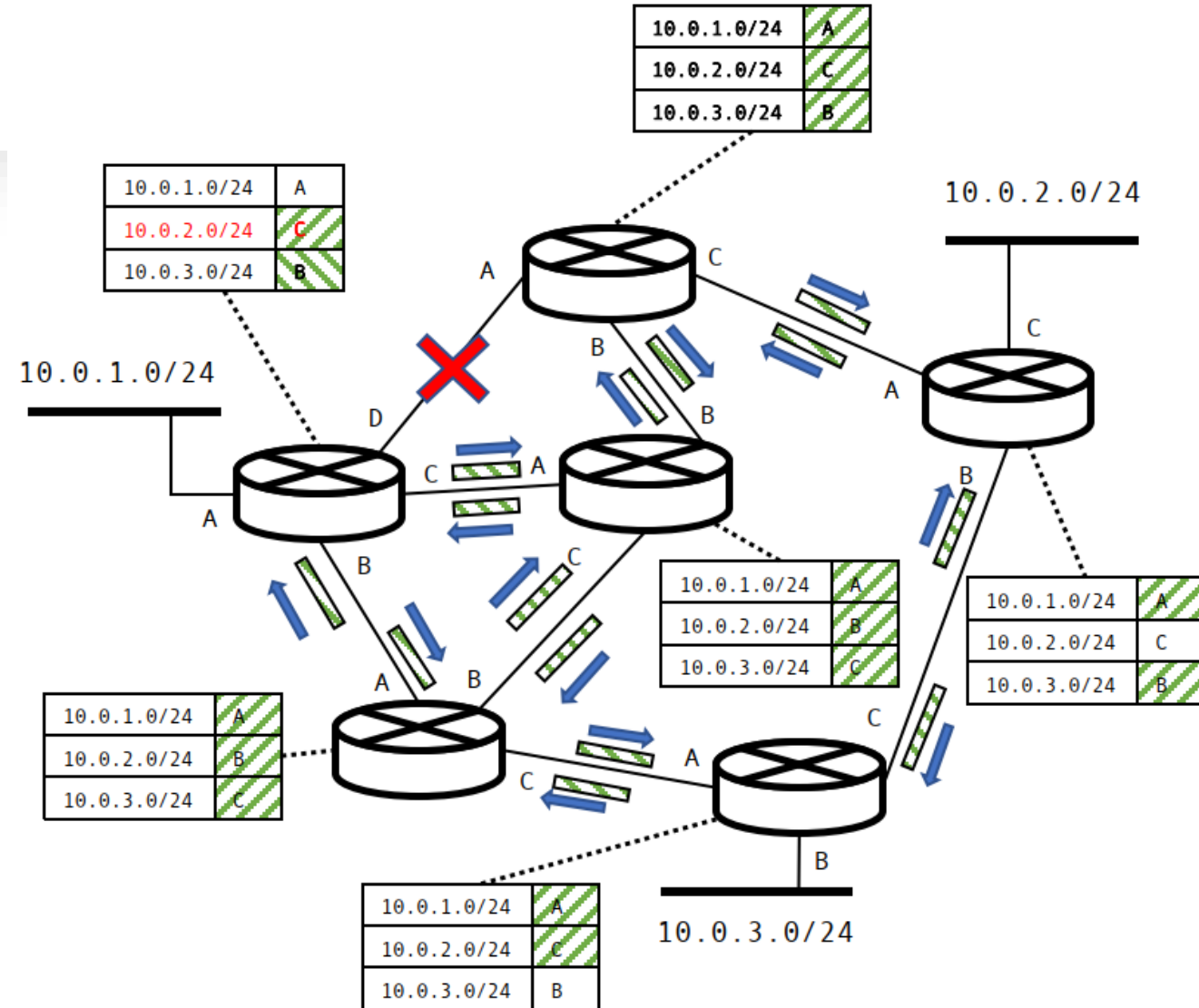
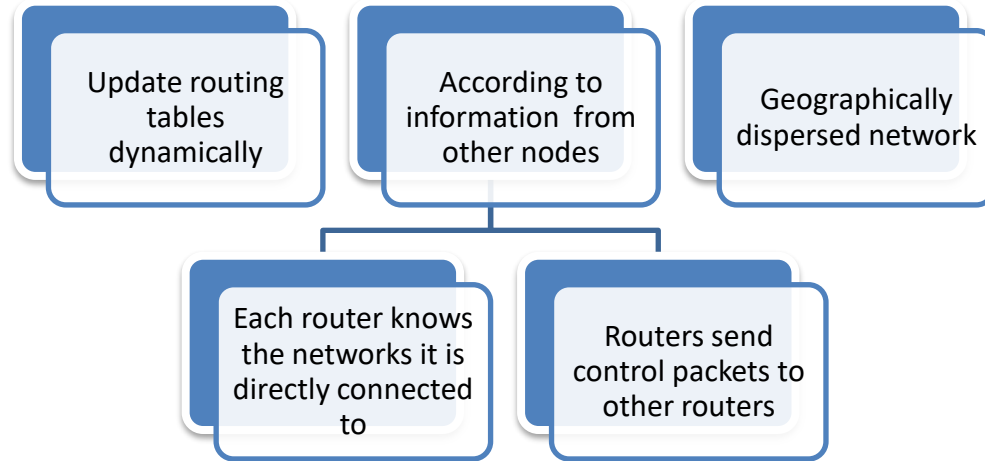
Returns routing table to each router

SDN-like solution

- Data centers



Centralized routing



Routing algorithms



Link state

Full topology of the network, then Dijkstra

Distributed: flood link state announcements – OSPF



Distance vector

Distance to destination node, Bellman-Ford

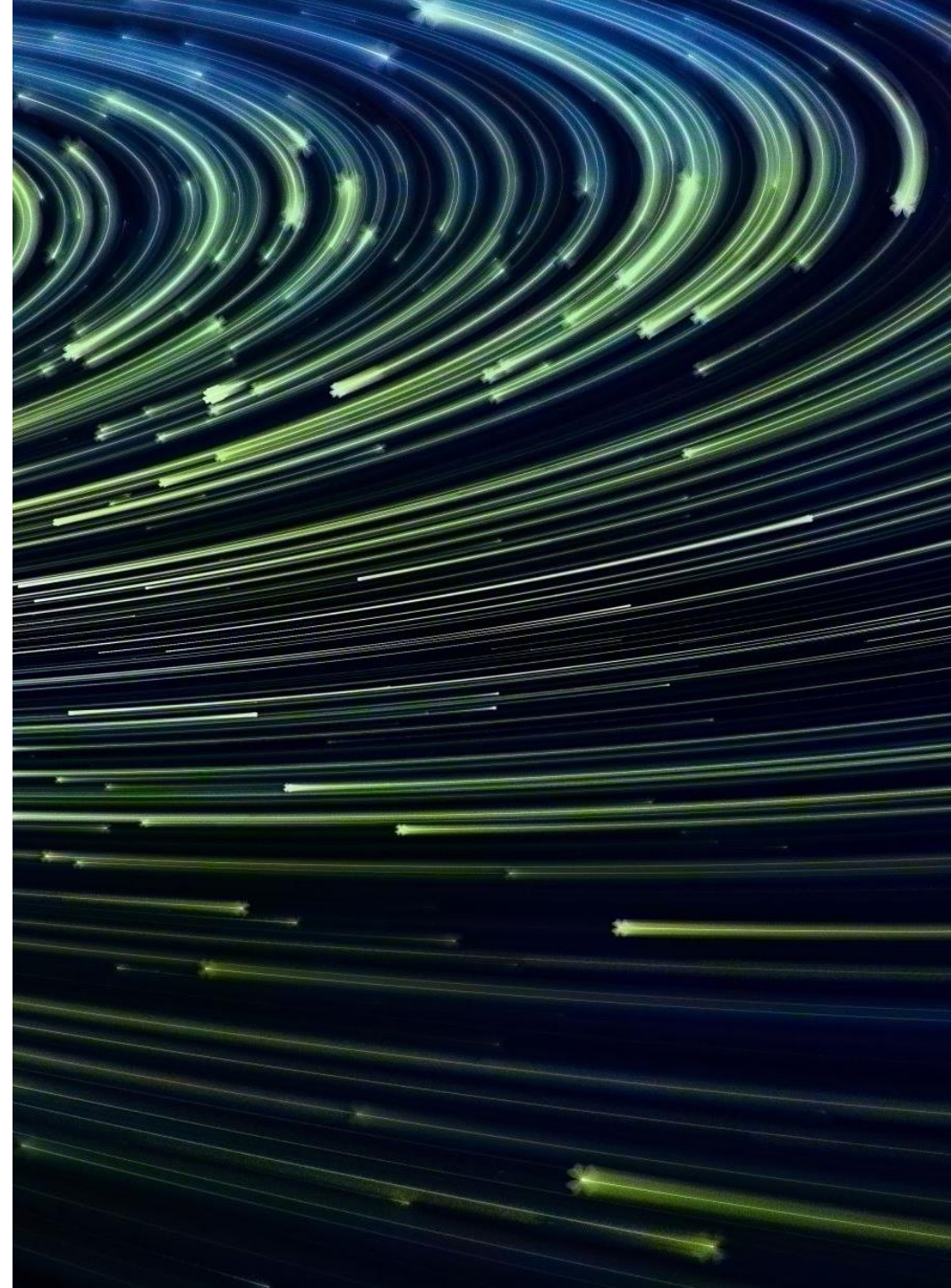
Distributed: update distances, resend to neighbors



Path vector

List of nodes to destination node, local preferences

Distributed: update path vectors, resend to neighbors



Interior routing

Same
administrative
domain

**All networks and devices
administered by same entity**

**Typically, the internal network of an
organization**

Protocols:

RIP (outdated) – distance vector

**EIGRP (Cisco) – distance vector,
optimized, hybrid**

OSPF – link state

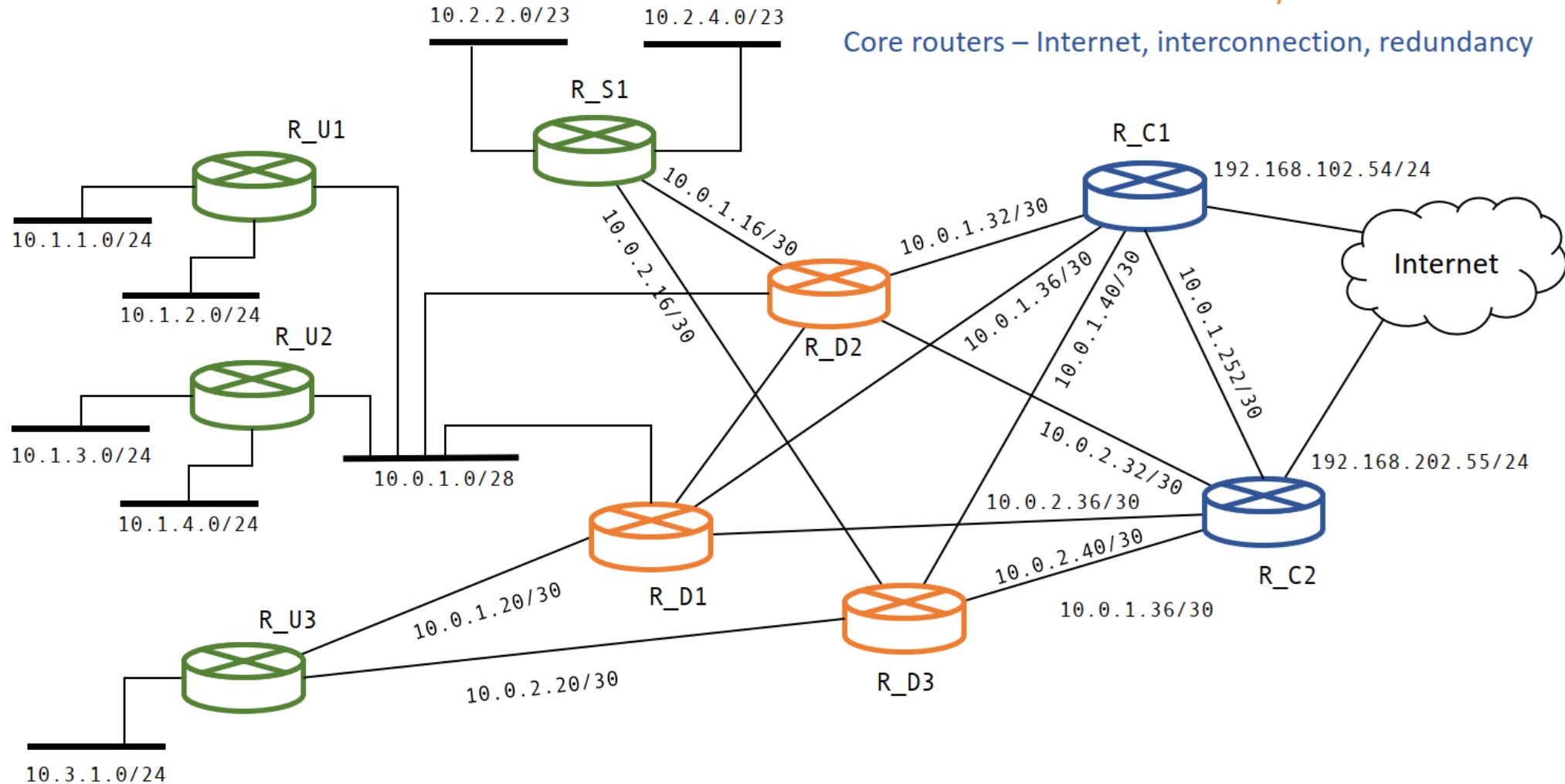


Interior routing

Access routers – end user traffic to Internet and servers

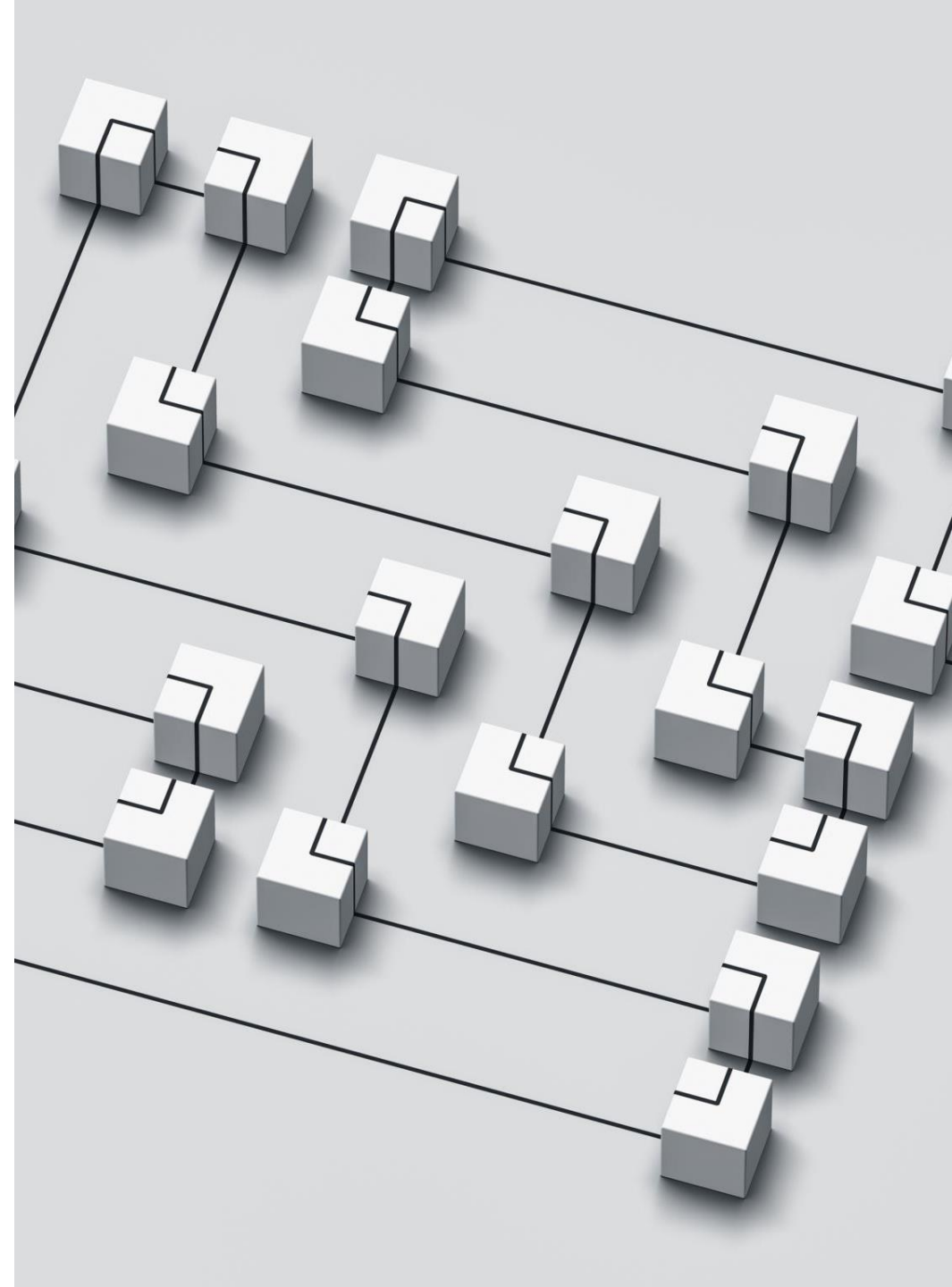
Distribution routers -- redundancy

Core routers – Internet, interconnection, redundancy



Introduction to OSPF (Open Shortest Path First)

- **OSPF Overview:**
 - Link-state routing protocol
 - Uses Dijkstra's Shortest Path First (SPF) algorithm
- **OSPF Characteristics:**
 - Hierarchical structure (Areas and Backbone Area)
 - Fast convergence
 - Supports VLSM and CIDR



How OSPF Works

- **OSPF Process:**
 - Neighbor Discovery
 - Database Exchange
 - SPF Calculation
 - Routing Table Update
- **LSAs (Link-State Advertisements):**
 - Used for topology information exchange



OSPF



<https://www.youtube.com/watch?v=kfvJ8QVJscc>

Exterior routing

Different
administrative
domains

**Each node in this network has a
different administrator**

The Internet

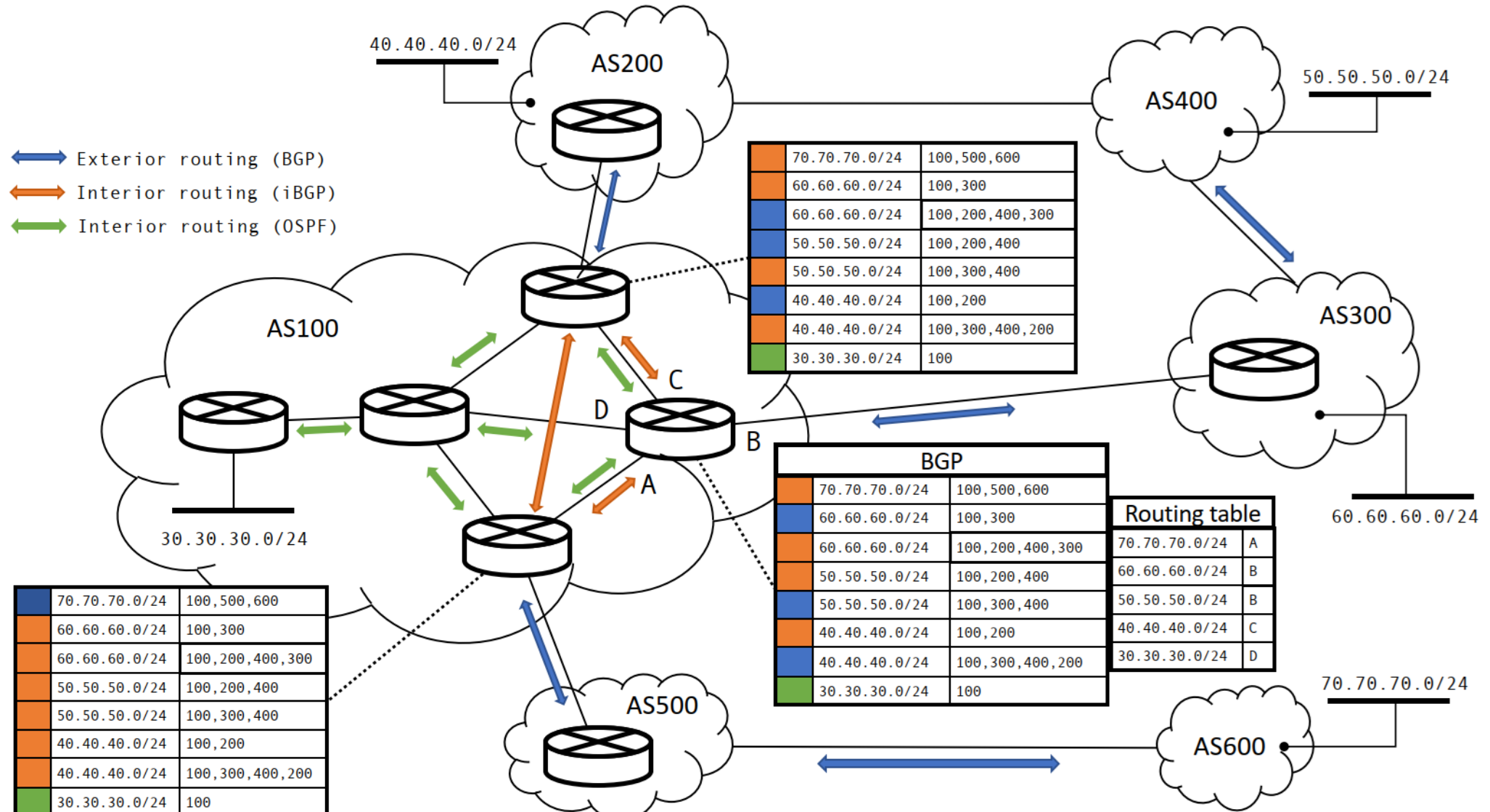
**Each node is an Autonomous System
– and a network by itself**

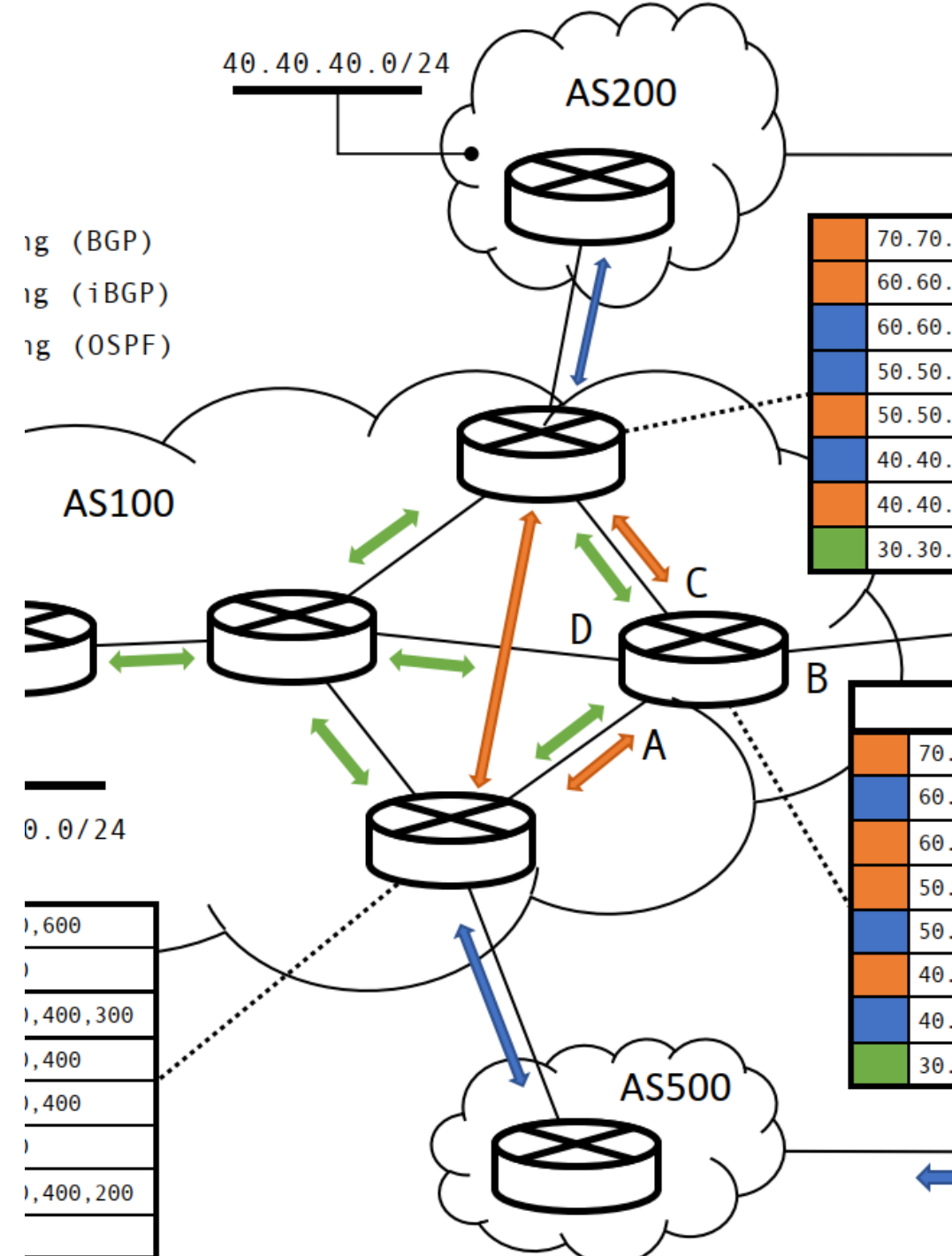
Protocols

BGP – path vector



Exterior routing



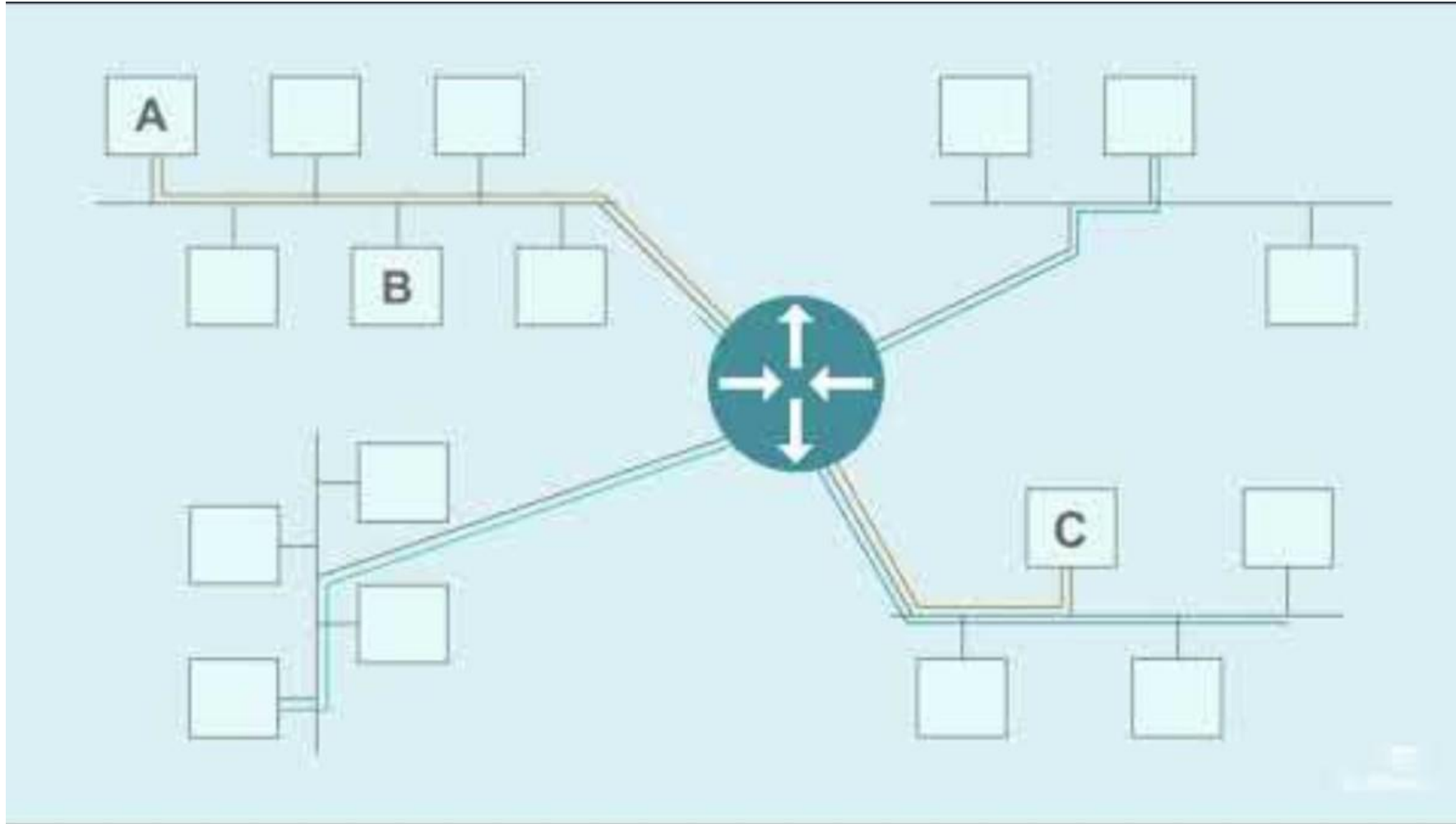


How BGP Works

- **Path Selection Criteria:**
 - Highest Local Preference
 - Shortest AS Path
 - Lowest MED (Multi-Exit Discriminator)
 - Prefer eBGP over iBGP
 - Lowest Router ID
- **BGP Attributes:**
 - AS Path, Next Hop, Local Preference, MED, Weight



BGP



<https://www.youtube.com/watch?v=A1KXPpqINZ4>

OSPF vs. BGP Comparison

Feature	OSPF	BGP
Type	Link-state	Path-vector
Use Case	Intra-domain (LAN)	Inter-domain (WAN, Internet)
Algorithm	Dijkstra (SPF)	Best Path Selection (Path Attributes)
Convergence Speed	Fast	Slow
Scalability	Medium	High

Use Cases & Real-World Applications

OSPF: Used in large
enterprise networks

Data centers, ISPs
(for internal routing)

BGP: Used for Internet
routing

Connects ISPs and
large organizations

Key Takeaways

OSPF is best for internal routing within organizations

BGP is essential for routing between ISPs and global networks

Both protocols play crucial roles in networking infrastructure

Choosing the right protocol depends on network size and scope

Further Reading & References

- **Books:**
 - "Routing TCP/IP, Volume 1" by Jeff Doyle
 - "Internet Routing Architectures" by Sam Halabi
- **Online Resources:**
 - Cisco OSPF and BGP Configuration Guides
 - RFC 2328 (OSPFv2) and RFC 4271 (BGP-4)
 - Networking blogs and forums (e.g., Packet Pushers, NetworkLessons)
- **Hands-on Practice:**
 - Use GNS3 or Cisco Packet Tracer to simulate OSPF and BGP networks

